SGA 7-I, Exposé VI

Source pages 121–137: Formal Jacobian subschemes and quadratic singularities

5. Formal Jacobian subschemes, continued

We use the notation introduced above for the Fitting ideals $I^p(E)$. For a morphism

$$f : X \longrightarrow Y$$

which is essentially of finite presentation, we denote by

$$I^p(X \mid Y) = I^p_X(\Omega^1_{X/Y})$$

the corresponding ideal sheaf on X, and by $J^p(X \mid Y)$ the closed subscheme which it defines. Thus $J^0(X \mid Y) \supset J^1(X \mid Y) \supset \cdots$.

Theorem 5.3. Let $f : X \to Y$ be essentially of finite presentation. Then:

(1) For every base change $Y' \to Y$, with $X' = X \times_Y Y'$, the canonical morphism

$$J^p(X' \mid Y') \longrightarrow J^p(X \mid Y) \times_Y Y'$$

is an isomorphism, for every p.

(2) For every $x \in X$, $y = f(x)$, the canonical local comparison is an isomorphism:

$$I^p_X(\Omega^1_{X/Y})_x \simeq I^p(\Omega^1_{\mathcal{O}_{X,x}/\mathcal{O}_{Y,y}}).$$

(3) The $J^p(X \mid Y)$ form a decreasing sequence of closed subschemes. A point x does not belong to $J^p(X \mid Y)$ once p is larger than

$$\dim_{\kappa(x)}(\Omega^1_{X/Y} \otimes \kappa(x)).$$

(4) One has $x \notin J^p(X \mid Y)$ if and only if, after replacing X by an open neighbourhood U of x, there exists a closed immersion

$$i : U \hookrightarrow U'$$

into a Y-scheme U' which is smooth over Y at $x' = i(x)$, and whose relative embedding dimension at x' is p.

(5) If X_1 and X_2 are two Y-schemes essentially of finite presentation and $X = X_1 \times_Y X_2$, then

$$I^p(X \mid Y) = \sum_{p_1+p_2=p} \mathrm{pr}_1^* I^{p_1}(X_1 \mid Y)\ \mathrm{pr}_2^* I^{p_2}(X_2 \mid Y).$$

Proof. Assertion (3) is the elementary monotonicity of Fitting ideals. Assertion (1) follows from the base-change formula for Fitting ideals and from

$$\Omega^1_{X'/Y'} \simeq \mathrm{pr}^* \Omega^1_{X/Y}.$$

Assertion (2) is the same statement after localization at x. Assertion (5) follows from the direct-sum formula for Fitting ideals applied to

$$\Omega^1_{X_1 \times_Y X_2/Y} \simeq \mathrm{pr}_1^* \Omega^1_{X_1/Y} \oplus \mathrm{pr}_2^* \Omega^1_{X_2/Y}.$$

For (4), suppose first that such an immersion $U \hookrightarrow U'$ exists. Since U' is smooth over Y at x', $\Omega^1_{U'/Y,x'}$ is free of rank p, and the exact conormal sequence

$$I/I^2 \longrightarrow \Omega^1_{U'/Y} \otimes \mathcal{O}_U \longrightarrow \Omega^1_{U/Y} \longrightarrow 0$$

shows that the p-th Fitting ideal is the unit ideal at x. Hence $x \notin J^p(U \mid Y)$, and therefore $x \notin J^p(X \mid Y)$.

Conversely, the question is local. Write $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and choose a presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \qquad P = A[x_1, \ldots, x_n].$$

The exact sequence

$$J/J^2 \longrightarrow B \otimes_P \Omega^1_{P/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

has middle term free of rank n. The condition $x \notin J^p(X \mid Y)$ means that a suitable $(n-p)$-minor of the Jacobian matrix of chosen generators of J is invertible at x. Thus there are elements

$$f_1, \ldots, f_{n-p} \in J$$

such that

$$U' = \operatorname{Spec}\big(P/(f_1, \ldots, f_{n-p})\big)$$

is smooth over Y at the image x' of x, and $U = \operatorname{Spec} B$ embeds as a closed subscheme of U'. This gives the required local presentation.

The following result is the corresponding Jacobian criterion.

Theorem 5.4. Let $f : X \to Y$ be essentially of finite presentation. Suppose that there is an integer $n \geq 0$ such that either:

(1) f is flat and all fibers have dimension n;

(2) f is a relative complete intersection of virtual dimension n in the sense of SGA 6, VIII 1.9.

Then

$$x \notin J^n(X \mid Y) \quad \Longleftrightarrow \quad f \text{ is smooth at } x.$$

Proof. By Theorem 5.3(4), $x \notin J^n(X \mid Y)$ if and only if locally X embeds into a Y-scheme U', smooth over Y at x', with relative dimension n. It remains to see that such an embedding is an isomorphism at x.

Let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be the corresponding exact sequence of local rings, with B' the local ring of U' at x' and B the local ring of X at x. Put $A = \mathcal{O}_{Y,y}$. In case (1), flatness gives the exact sequence on the closed fiber

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0,$$

where $\mathfrak{m}$ is the maximal ideal of A. The two fiber local rings have the same dimension n, and $B'/\mathfrak{m}B'$ is regular; hence $I/\mathfrak{m}I = 0$, whence $I = 0$.

In case (2), the conormal sequence

$$I/I^2 \longrightarrow B \otimes_{B'} \Omega^1_{B'/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

is exact. Since X is a relative complete intersection and U' is smooth, the first map is injective and the two modules which occur have the ranks prescribed by the virtual dimension. These ranks force $I/I^2 = 0$, hence $I = 0$.

Definition 5.5. Under the hypotheses of Theorem 5.4, one writes simply

$$I(X \mid Y) = I^n(X \mid Y), \qquad J(X \mid Y) = J^n(X \mid Y).$$

Thus $J(X \mid Y)$ is the Jacobian subscheme measuring failure of smoothness.

We now extend the construction to noetherian formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and E a coherent $\mathcal{O}_{\mathfrak{X}}$-module. Define

$$I^p_{\mathfrak{X}}(E)(U) = I^p(\Gamma(U, E))$$

for every affine open $U \subset \mathfrak{X}$. If $\mathfrak{X} = \varprojlim X_v$ and $E = \varprojlim E_v$, then

$$I^p_{\mathfrak{X}}(E) = \varprojlim_v I^p_{X_v}(E_v).$$

Let S be a noetherian formal scheme and let $\mathfrak{X}$ be an S-adic formal scheme, essentially of finite type. Let

$$\mathfrak{X} = \varprojlim X_v, \qquad S = \varprojlim S_v, \qquad X_v = \mathfrak{X} \times_S S_v.$$

Then

$$\Omega^1_{\mathfrak{X}/S} \simeq \varprojlim_v \Omega^1_{X_v/S_v}$$

is a coherent $\mathcal{O}_{\mathfrak{X}}$-module and is compatible with finite-type formal base change.

Definition 5.6. Define

$$I^p(\mathfrak{X} \mid S) = I^p_{\mathfrak{X}}(\Omega^1_{\mathfrak{X}/S})$$

and let $J^p(\mathfrak{X} \mid S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal. Equivalently

$$I^p(\mathfrak{X} \mid S) = \varprojlim_v I^p(X_v \mid S_v), \qquad J^p(\mathfrak{X} \mid S) = \varprojlim_v J^p(X_v \mid S_v).$$

There is the natural cartesian comparison with the closed fiber:

$$\begin{array}{ccc} J^p(X_0 \mid S_0) & \longrightarrow & J^p(\mathfrak{X} \mid S) \\ \downarrow & & \downarrow \\ X_0 & \longrightarrow & \mathfrak{X}. \end{array}$$

If x is an isolated point of $J^p(\mathfrak{X} \mid S)$, then the formal completion of $J^p(\mathfrak{X} \mid S)$ at x is canonically the corresponding connected component. In particular, if

$$|J^p(X_0 \mid S_0)| = \{x_1, \dots, x_m\}$$

is finite and discrete, then

$$J^p(\mathfrak{X} \mid S) \simeq \coprod_{1 \le i \le m} J^p(\mathfrak{X} \mid S)_{(x_i)}.$$

Corollary 5.7. Let S be noetherian and formal, and let $\mathfrak{X}$ be an S-adic formal scheme essentially of finite type.

(1) The construction $J^p(\mathfrak{X} \mid S)$ commutes with formal base change:

$$J^p(\mathfrak{X}' \mid S') = J^p(\mathfrak{X} \mid S) \times_S S', \qquad \mathfrak{X}' = \mathfrak{X} \times_S S'.$$

(2) The connected components of $J^p(\mathfrak{X} \mid S)$ are in bijection with those of $J^p(X_0 \mid S_0)$, and for each isolated point x of $J^p(X_0 \mid S_0)$ there is a canonical isomorphism of completed local pieces

$$J^p(\mathfrak{X} \mid S)_{(x)} \simeq J^p(X_0 \mid S_0)_{(x)}.$$

In particular, in the finite discrete case one has the above disjoint-union decomposition.

(3) If S is an ordinary noetherian scheme, $S_0 \subset S$ a closed subscheme, $\widehat{S}$ the formal completion of S along S_0, and

$$\widehat{X} = X \times_S \widehat{S},$$

then

$$J^p(\widehat{X} \mid \widehat{S}) = J^p(X \mid S) \times_S \widehat{S}.$$

Thus the formal Jacobian subscheme is the completion of $J^p(X \mid S)$ along $J^p(X_0 \mid S_0)$.

(4) If a fixed integer n satisfies the hypotheses of Theorem 5.4 for all $X_v \to S_v$, then $x \notin J^n(\mathfrak{X} \mid S)$ if and only if $\mathfrak{X}$ is formally smooth over S at x.

Proof. The assertions follow directly from the corresponding statements for ordinary schemes, the compatibility of $\Omega^1_{\mathfrak{X}/S}$ with passage to X_v/S_v, and the base-change formula for Fitting ideals.

Remark 5.8. Let $\mathfrak{X}$ be S-adic, essentially of finite type, and let $x \in \mathfrak{X}$ with image $y \in S$. One can also consider the local formal $S_{(y)}$-scheme

$$\mathfrak{X}_{(x)} = \varprojlim_v \operatorname{Spec}(\mathcal{O}_{X_v,x_v}),$$

with the natural morphism $\mathfrak{X}_{(x)} \to \mathfrak{X}$. Although $\mathfrak{X}_{(x)}$ need not itself be essentially of finite type over $S_{(y)}$, the completed module of differentials is still coherent in the sense needed here:

$$\Omega^1_{\mathfrak{X}_{(x)}/S_{(y)}} \simeq \varprojlim_v \widehat{\Omega^1_{\mathcal{O}_{X_v,x_v}/\mathcal{O}_{S_v,y_v}}}.$$

This gives local formal subschemes $J^p(\mathfrak{X}_{(x)} \mid S)$, and if x is an isolated point of $J^p(\mathfrak{X} \mid S)$, the natural map

$$J^p(\mathfrak{X}_{(x)} \mid S) \longrightarrow J^p(\mathfrak{X} \mid S)$$

identifies the source with the completed local component at x.

We shall also need the image of the Jacobian subscheme in the base. The set-theoretic image may be described by kernels

$$\operatorname{Ker}\big(\mathcal{O}_S \to f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big),$$

but the Fitting-ideal formulation is more useful.

Definition 5.9. Let S be noetherian and formal, and let

$$f : \mathfrak{X} \longrightarrow S$$

be S-adic, essentially of finite type. Suppose $J^p(\mathfrak{X} \mid S)$ is proper over S. Then $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$ is coherent, and we define an ideal sheaf on S by

$$I^p_S(\mathfrak{X} \mid S) = I^0_S\big(f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big).$$

Let $J^p_S(\mathfrak{X} \mid S)$ be the closed formal subscheme of S defined by this ideal. If the fixed-dimension hypotheses of Theorem 5.4 are in force, one writes

$$I_S(\mathfrak{X} \mid S) = I^n_S(\mathfrak{X} \mid S), \qquad J_S(\mathfrak{X} \mid S) = J^n_S(\mathfrak{X} \mid S).$$

Corollary 5.10. Under the hypotheses of Definition 5.9:

(1) The underlying set of $J^p_S(\mathfrak{X} \mid S)$ is the image of $J^p(\mathfrak{X} \mid S)$ under $f : \mathfrak{X} \to S$.

(2) The formation of $J^p_S(\mathfrak{X} \mid S)$ commutes with base change.

(3) If
$$J^p(\mathfrak{X} \mid S) = \coprod_i J^p(\mathfrak{X} \mid S)_{(x_i)}$$
is the disjoint union of its isolated formal local components, then
$$I^p_S(\mathfrak{X} \mid S) = \prod_i I^p_{S,(x_i)}(\mathfrak{X} \mid S).$$

(4) In the case of completion of an ordinary noetherian S-scheme X, the formal closed subscheme $J^p_S(\widehat{X} \mid \widehat{S})$ is the completion of $J^p_S(X \mid S)$.

Proof. Assertion (1) is the usual support property of the zeroth Fitting ideal. Assertions (2) and (4) follow from base change for Fitting ideals. Assertion (3) follows from the product formula for I^0 applied to the decomposition of the coherent algebra $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$.

Remark 5.11. The Jacobian ideal above is classical: it commutes with base change and, under the mild dimension hypotheses of Theorem 5.4, gives the smoothness criterion. There are also more intrinsic ways of defining a canonical non-smooth locus in complete generality. For an A-algebra B, set
$$I_{B/A} = \bigcap_E \operatorname{Ann}_B D^1(B \mid A, E),$$
where E runs through the finite-type B-modules. This ideal commutes with localization by Corollary 3.13(c). Hence for a morphism $X \to Y$ one obtains an ideal sheaf $I_{X/Y}$. If $X \to Y$ is essentially of finite type, then f is smooth at x if and only if
$$x \notin \operatorname{Supp}(\mathcal{O}_X / I_{X/Y})$$
by 3.13(a). The same construction carries over to adic formal schemes essentially of finite type and is therefore also suited to deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k-scheme whose singularities are isolated non-degenerate quadratic singularities. In view of 4.13, under the hypotheses considered here the question becomes a local one.

We return to the notation of Section 4. Thus A is a fixed complete noetherian local ring with residue field k, and $\mathcal{C}_A$ is the category of local artinian A-algebras with residue field k. Let X be a k-scheme of finite type and let x be a closed point of X. Recall that x is a non-degenerate quadratic singularity, rational over k, if
$$\widehat{\mathcal{O}}_{X,x} \simeq k[[x_1, \dots, x_n]]/(f)$$
and
$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = (x_1, \dots, x_n).$$
One may then arrange that f itself is a quadratic form.

Lemma 6.1. Let
$$f \in k[[x_1, \dots, x_n]]$$

and suppose

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = \mathfrak{m} = (x_1, \dots, x_n).$$

Then there are parameters $x'_1, \dots, x'_n$, with

$$x_i \equiv x'_i \pmod{\mathfrak{m}^2},$$

such that

$$f = q(x'_1, \dots, x'_n)$$

for a quadratic form q.

Proof. Write $f = f_2 + h_1$, where f_2 is quadratic and $h_1 \in \mathfrak{m}^3$. Since the partial derivatives generate $\mathfrak{m}$, there are elements $e_i^{(1)} \in \mathfrak{m}^2$ such that

$$\sum_i e_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_1 \pmod{\mathfrak{m}^4}.$$

After replacing x_i by $x_i - e_i^{(1)}$, the remainder is pushed into $\mathfrak{m}^4$. Repeating this construction, one obtains $e_i^{(v)} \in \mathfrak{m}^{v+1}$ and parameters

$$x'_i = x_i - \sum_{v \geq 1} e_i^{(v)}$$

for which all terms of degree at least 3 disappear. Thus f becomes its quadratic part in the new variables.

Lemma 6.2. Let

$$A_0 \simeq k[[x_1, \dots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then:

(1) The natural maps

$$J(\mathfrak{X} \mid R) \longrightarrow J_R(\mathfrak{X} \mid R) \longrightarrow \operatorname{Spec} A$$

are isomorphisms.

(2) The base Jacobian ideal is principal:

$$I_R(\mathfrak{X} \mid R) = (t), \qquad R = A[[t]],$$

where t is a free formal generator of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. By 4.10(b), the formal versal deformation has the explicit form

$$R = A[[t]], \qquad \mathfrak{X} = \operatorname{Spf}\big(R[[x_1, \dots, x_n]]/(F - t)\big),$$

where F is obtained from f by lifting the coefficients to A. The exact sequence

$$A_0 \xrightarrow{dF} \bigoplus_i A_0 \, dx_i \longrightarrow \Omega^1_{A_0/R} \longrightarrow 0$$

shows that

$$I(\mathfrak{X} \mid R) = (\partial F/\partial x_1, \dots, \partial F/\partial x_n) = (x_1, \dots, x_n)$$

on the special fiber. Hence the Jacobian subscheme is obtained by setting $x_1 = \cdots = x_n = 0$, and the remaining equation is $t = 0$. Thus the maps in (1) are isomorphisms and $I_R(\mathfrak{X} \mid R) = (t)$. Finally

$$R[[x_1, \dots, x_n]]/(F - t) \simeq A[[x_1, \dots, x_n]]$$

as an A-algebra, proving formal smoothness over A.

Corollary 6.3. Let $x \in X_0$ be a non-degenerate quadratic singularity rational over k.

(1) Let $(R, \mathfrak{X}_1)$ and $(R, \mathfrak{X}_2)$ be two deformations of X_0 over an object R of $\mathcal{C}_A$, with the same point above x. Then the completed local R-adic formal schemes $(\mathfrak{X}_1)_{(x)}$ and $(\mathfrak{X}_2)_{(x)}$ are isomorphic over R if and only if

$$I_R((\mathfrak{X}_1)_{(x)} \mid R) = I_R((\mathfrak{X}_2)_{(x)} \mid R).$$

(2) Let R be a regular local ring and $(R, \mathfrak{X})$ a deformation of X_0. Then $\mathcal{O}_{\mathfrak{X},x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)} \mid R) \not\subset \mathfrak{m}_R^2,$$

that is, if and only if this Jacobian ideal defines a regular divisor in $\operatorname{Spf} R$.

Proof. Let the formal versal deformation be

$$A[[t]][[x_1, \dots, x_n]]/(F - t).$$

Two induced deformations over R are given by two elements $a, b \in R$:

$$R[[x_1, \dots, x_n]]/(F - a), \qquad R[[x_1, \dots, x_n]]/(F - b).$$

Their Jacobian ideals are (a) and (b). Equality of these ideals gives $a = vb$ for a unit v. Since R is complete and v has a square root after the standard lifting argument used here, the change of variables $x_i \mapsto v^{1/2} x_i$ identifies the two equations, because F is quadratic. The converse is immediate. The second assertion follows from the fact that the quotient of the regular local ring $R[[x_1, \dots, x_n]]$ by $F - a$ is regular precisely when $a \notin \mathfrak{m}_R^2$.

Theorem 6.4. Let X_0 be a k-scheme of finite type with only isolated non-smooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Let

$$\{z_1, \dots, z_m\}$$

be the set of non-smooth points of X_0. Suppose

$$H^2(X_0, D^0) = 0$$

and suppose that all z_i are non-degenerate quadratic singularities, rational over k. Then:

(1) R is formally smooth over A, i.e. R is a formal power-series ring over A.

(2) The Jacobian subscheme decomposes as a disjoint union

$$J(\mathfrak{X} \mid R) = \coprod_{i=1}^{m} J(\mathfrak{X}_{(z_i)} \mid R),$$

and for each i the natural maps from the local Jacobian component to its image in $\operatorname{Spf} R$ are isomorphisms.

(3) The ideal $I_R(\mathfrak{X}_{(z_i)} \mid R)$ is principal. If

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

then $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over A, and

$$I_R(\mathfrak{X} \mid R) = \prod_{i=1}^{m} (t_i).$$

(4) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(5) If X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

Proof. For each singular point z_i, let $(R_i, \mathfrak{U}_i)$ be the formal versal deformation of $\operatorname{Spec}(\mathcal{O}_{X_0,z_i})$. Since $(R, \mathfrak{X})$ induces a deformation of this local scheme, there is a continuous local map $R_i \to R$ and a cartesian diagram

$$\begin{array}{ccc} \mathfrak{X}_{(z_i)} & \longrightarrow & \mathfrak{U}_i \\ \downarrow & & \downarrow \\ \operatorname{Spf} R & \longrightarrow & \operatorname{Spf} R_i. \end{array}$$

Lemma 6.2 gives

$$J(\mathfrak{X}_{(z_i)} \mid R) \simeq J(\mathfrak{U}_i \mid R_i) \times_{\operatorname{Spf} R_i} \operatorname{Spf} R$$

and

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

where t_i is the image in R of the corresponding local parameter. Since $H^2(X_0, D^0) = 0$, the local-to-global criterion of 4.13 implies that

$$\operatorname{Spf} R \longrightarrow \prod_i \operatorname{Spf} R_i$$

is formally smooth in the complementary directions; equivalently $t_1, \ldots, t_m$ extend to a system of free formal generators of R. This proves (1), (2), and (3). Assertion (4) follows because each local singular piece is formally smooth over $\operatorname{Spec} A$ by Lemma 6.2(3), while away from the z_i the original scheme is already smooth.

For (5), formal smoothness of R over A, together with the exact sequence obtained in 4.4 and 4.11,

$$0 \to H^1(X_0, D^0) \to \mathfrak{m}_R/\mathfrak{m}_R^2 \to H^0(X_0, D^1) \to 0,$$

gives

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = \dim_k H^1(X_0, D^0) + \dim_k H^0(X_0, D^1).$$

Thus it remains to compute the Euler characteristic of D^0 on the curve. After extension of the base field to an algebraic closure, each non-degenerate quadratic singularity of a curve has completed local ring

$$k[[x, y]]/(xy).$$

If $A = \mathcal{O}_{X_0,z}$ and A' is its normalization, then

$$\ell(A'/A) = 1, \qquad \operatorname{Der}_k(A, \mathfrak{m}_A) = \operatorname{Der}_k(A, A),$$

and the conductor C of A in A' is $\mathfrak{m}_A$. Consequently

$$\operatorname{Der}_k(A, A) = \operatorname{Der}_k(A', A')$$

as A-submodules of $\operatorname{Der}_k(\operatorname{Frac} A, \operatorname{Frac} A)$, and one obtains an exact sequence

$$0 \to D^0_{X_0} \to D^0_{\widetilde{X}_0} \to \mathcal{O}_{\widetilde{X}_0}/C \to 0,$$

where $\widetilde{X}_0$ denotes the normalization. Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{\widetilde{X}_0}/C) = 2m.$$

Riemann–Roch on the normalization gives

$$\chi(D^0_{\widetilde{X}_0}) = 3 - 3(g - m).$$

Therefore

$$\chi(D^0_{X_0}) = \chi(D^0_{\widetilde{X}_0}) - \ell(\mathcal{O}_{\widetilde{X}_0}/C) = 3 - 3g + m,$$

which is equivalent to the stated formula.

Remark 6.5. In the formula of Theorem 6.4(5), the integer

$$a = \dim_k H^0(X_0, D^1)$$

is easily computed by Riemann–Roch. Namely

$$a = \begin{cases} 0, & g \geq 2, \\ 1, & g = 1, \\ 3, & g = 0. \end{cases}$$

Finally, in characteristic 2 there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension when $\operatorname{char} k = 2$. In that case one uses quadratic forms of maximum possible non-degeneracy.

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity, rational over k, if:

(i) either $\operatorname{char} k \neq 2$, or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd; in this case z is required to be a non-degenerate quadratic singularity rational over k;

(ii) or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is even; after extension to an algebraic closure $\bar{k}$, the completed local ring has the form

$$\bar{k}[[x_0, x_1, \ldots, x_{2n}]]/(x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}).$$

Lemma 6.7. Assume $\operatorname{char} k = 2$, and put

$$A_0 = k[[x_0, x_1, \ldots, x_{2n}]]/(f), \qquad f = x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then

$$R = A[[t_1, t_2]],$$

where t_1, t_2 are free formal generators over A,

$$I_R(\mathfrak{X} \mid R) = (t_1^2),$$

and $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. Since

$$\left(\frac{\partial f}{\partial x_0}, \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_{2n}}\right) = (x_2, x_1, \dots, x_{2n}, x_{2n-1}),$$

the deformation space has two formal parameters. By 4.14 the versal deformation may be written

$$R = A[[t_1, t_2]], \qquad \mathfrak{X} = \operatorname{Spf}(R[[x_0, \dots, x_{2n}]]/(F)),$$

with

$$F = f + t_1 + t_2 x_0.$$

The exact sequence for differentials gives

$$I(\mathfrak{X} \mid R) = (t_2, x_1, \dots, x_{2n}),$$

and the residual equation on the Jacobian locus is t_1. Thus the induced base ideal is (t_1^2). Moreover the equation can be rewritten so that the total formal scheme is formally smooth over A, giving the last assertion.

Corollary 6.8. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Suppose

$$H^2(X_0, D^0) = 0,$$

$\operatorname{char} k = 2$, $\dim X_0$ is even, and all non-smooth points $z_1, \dots, z_m$ are ordinary quadratic singularities. Then:

(1) R is formally smooth over A.

(2) Each local base Jacobian ideal is principal of the form

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i^2),$$

and

$$I_R(\mathfrak{X} \mid R) = \prod_i (t_i^2).$$

Moreover $t_1, \dots, t_m$ can be completed to a system of free formal generators of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

The proof is the same as that of Theorem 6.4, using Lemma 6.7 in place of Lemma 6.2 for the local calculation in characteristic 2.